



Generate Footprint Points

Export the sidecut as tip, RSL and tail point tables at equal arc-length spacing, each relative to its own origin -- for CNC, specs and spreadsheets.

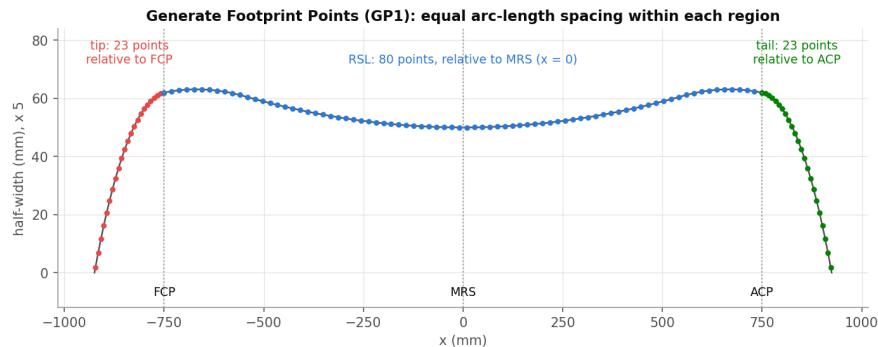


What it does

Shops and spec sheets want numbers, not curves. Generate Footprint Points splits the +Y outline at the contacts and samples each region evenly, in a table you can copy out.

Use it when

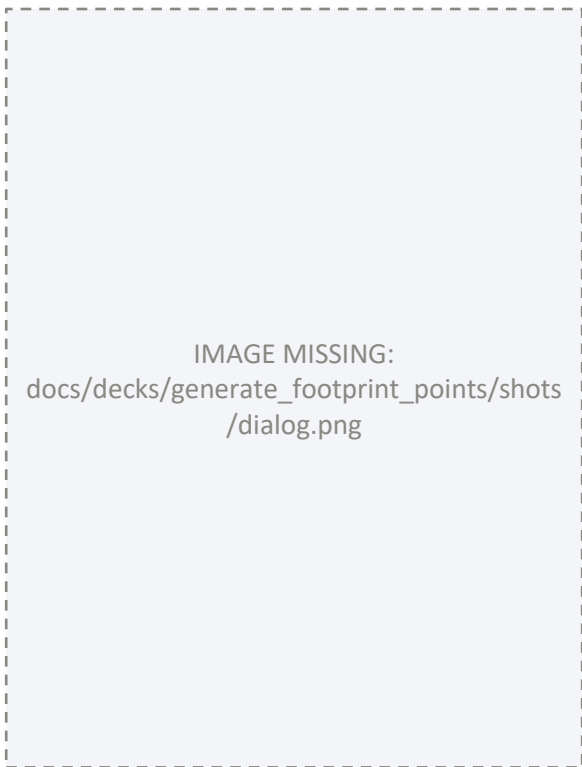
- Send a sidecut to CNC or a supplier
- Fill a spec sheet
- Compare outlines point by point



GP1: 23 tip, 80 RSL and 23 tail points on S14.



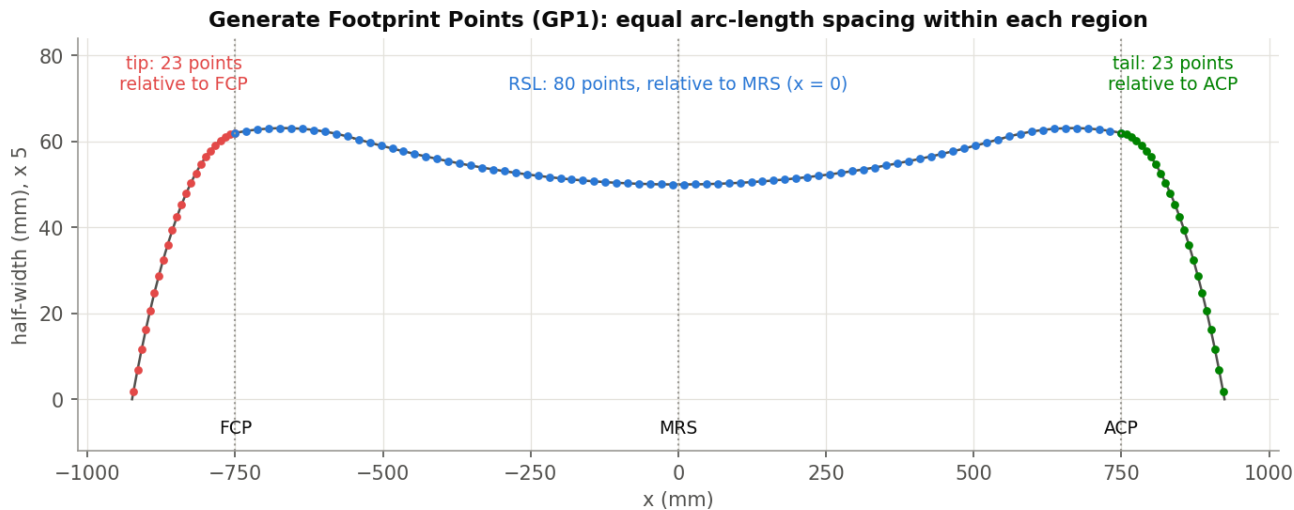
The dialog



#	Parameter	What it means
1	Geometry Selection Type	FACE (the footprint face) / EDGES (edges or wires).
2	Footprint Edges	(EDGES) the outline.
3	RSL Line	FCP to ACP.
4	Tip toward +X	Off: the tip is the lower-X end; on: the higher-X end.
5	Include Points	Not used.
6	Table Formatting	Units, decimal places ('sig figs'), units in cells, point counts 23 / 80 / 23, flips.
7	Additional options	Keep the region curves / FCP-ACP planes; make sketches of the points.



Example: GP1



S14, RSL -750 ... 750

Tip: 23 points x -924.567 ... -750 relative to the FCP; RSL: 80 points relative to the MRS; tail: 23 points relative to the ACP. All on the curve, spacing equal to 0.2 %.



Outputs and messages

Published for Extract variables

Key	What it holds
tipCurve, rslCurve, tailCurve	The region curves (empty unless Retain curves is on).
Footprint Points table	Reads tipEdgePoints / rslEdgePoints / tailEdgePoints / tableFormat stored on the origin (keep these stable).

Messages

	Message / what to do
(none)	



Tips and limits



Tip at +X

Turn on Tip toward +X; tip and tail swap with their origins (GP2).



Last one wins

The stored point lists are shared: the last Generate Footprint Points in the tree feeds the table.



Where it lives

Document	footprint (52724f3a857fa52d3ecceb77)
Tab	getFootprintPoints
Tests	Footprint tests studio (build_ / check_footprint_tests.py): GP1-GP3
Explainer	docs/explainers/footprint/footprint_explained.md, section 2.5
Related	Integrate footprint, Analyze footprint, Scale Footprint, Arc fit, Generate Footprint Points